

AI ASSISTED CODING

Assignment-15.1

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Task 1 – Student Records API

Task:

Use AI to build a RESTful API for managing student records.

Instructions:

- Endpoints required:
 - o GET /students → List all students
 - o POST /students → Add a new student
 - o PUT /students/{id} → Update student details
 - o DELETE /students/{id} → Delete a student record
- Use an in-memory data structure (list or dictionary) to store records.
- Ensure API responses are in JSON format.

Expected Output:

- Working API with CRUD functionality for student records.

Commands + Code + Text Run all



[] Start coding or generate with AI.

[1] `pip install fastapi uvicorn`

Requirement already satisfied: fastapi in /usr/local/lib/python3.12/dist-packages (0.135.1)
Requirement already satisfied: uvicorn in /usr/local/lib/python3.12/dist-packages (0.41.0)
Requirement already satisfied: starlette>=0.46.0 in /usr/local/lib/python3.12/dist-packages (from fastapi) (0.46.0)
Requirement already satisfied: pydantic>=2.7.0 in /usr/local/lib/python3.12/dist-packages (from fastapi) (2.10.6)
Requirement already satisfied: typing-extensions>=4.8.0 in /usr/local/lib/python3.12/dist-packages (from fastapi) (4.12.0)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from fastapi) (0.10.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from fastapi) (0.0.2)
Requirement already satisfied: click>=7.0 in /usr/local/lib/python3.12/dist-packages (from uvicorn) (8.1.8)
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.12/dist-packages (from uvicorn) (0.14.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic) (2.41.4)
Requirement already satisfied: anyio<5,>=3.6.2 in /usr/local/lib/python3.12/dist-packages (from anyio) (4.6.2)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio) (3.10)

I've installed `fastapi` and `uvicorn`. Now, let's define the API. I'll use `pydantic` for data validation and `fastapi` for the API endpoints. The student records will be stored in a simple Python dictionary in memory.

[2]
✓ 0s

```
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
from typing import Dict, List

app = FastAPI()

# In-memory data store for students
students_db: Dict[int, BaseModel] = {}
next_student_id = 1

# Pydantic model for a student
class Student(BaseModel):
    name: str
    age: int
    major: str

# Pydantic model for updating a student (all fields optional)
class StudentUpdate(BaseModel):
    name: str = None
    age: int = None
    major: str = None

@app.get("/students", response_model=List[Student])
def list_students():
```

```
[2] ✓ 0s
def list_students():
    """List all students."""
    return list(students_db.values())

@app.post("/students", response_model=Student, status_code=201)
def add_student(student: Student):
    """Add a new student."""
    global next_student_id
    students_db[next_student_id] = student
    student_with_id = student.copy(update={"id": next_student_id})
    next_student_id += 1
    return student_with_id

@app.put("/students/{student_id}", response_model=Student)
def update_student(student_id: int, student_update: StudentUpdate):
    """Update student details."""
    if student_id not in students_db:
        raise HTTPException(status_code=404, detail="Student not found")

    existing_student = students_db[student_id]
    update_data = student_update.dict(exclude_unset=True)
    updated_student = existing_student.copy(update=update_data)
    students_db[student_id] = updated_student
    return updated_student
```

```
@app.delete("/students/{student_id}", status_code=204)
def delete_student(student_id: int):
    """Delete a student record."""
    if student_id not in students_db:
        raise HTTPException(status_code=404, detail="Student not found")
    del students_db[student_id]
    return {"message": "Student deleted successfully"}
```

Now that the API is defined, you can run it using Uvicorn. Since you are in a Colab environment, we need to use `nest_asyncio` to allow Uvicorn to run within the existing event loop. Also, the API will be accessible via `localhost:8000`. You can test it by running this cell and then using `curl` or a tool like Postman/Insomnia in a separate terminal or web browser.

To interact with the API, you can use the following in a new cell or a separate terminal:

- **GET all students:**

```
!curl http://localhost:8000/students
```

- **POST a new student:**

```
!curl -X POST -H "Content-Type: application/json" -d '{"name": "Alice", "age": 20, "major": "Computer Science"}' http://localhost:8000/students
```

- **PUT (update) a student (assuming ID 1 exists):**

```
!curl -X PUT -H "Content-Type: application/json" -d '{"age": 21}' http://localhost:8000/students/1
```

- **DELETE a student (assuming ID 1 exists):**

```
!curl -X DELETE http://localhost:8000/students/1
```

The image consists of two screenshots of a Jupyter Notebook interface. The top screenshot shows the initial code for setting up a Uvicorn server. It imports `uvicorn`, `nest_asyncio`, and `asyncio`. It then uses `nest_asyncio.apply()` to allow Uvicorn to run with an asyncio loop. A configuration object is created for Uvicorn, and a server instance is initialized. A `ServerStopper` class is defined with an `__init__` method that takes the server instance and a `start` method that prints the API URL and starts the server. A `stop` method is also defined. The bottom screenshot shows the continuation of the code. It completes the `stop` method, which prints a message when the server is stopped. It then creates an instance of `ServerStopper`, runs the server in a separate task using `asyncio.create_task`, and provides instructions on how to start and stop the server. Finally, it calls `stopper.start()` to start the server. The bottom of the notebook shows the output of the code execution, indicating that the server process has started and is waiting for application startup, with the API URL printed.

```
[4]: import uvicorn
import nest_asyncio
import asyncio

nest_asyncio.apply()

# Configuration for Uvicorn server
config = uvicorn.Config(app, host="0.0.0.0", port=8000, loop="asyncio")
server = uvicorn.Server(config)

# Define a stopper to gracefully stop the server
class ServerStopper:
    def __init__(self, server_instance):
        self.server = server_instance
        self.task = None

    async def start(self):
        print("API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000")
        self.task = asyncio.create_task(self.server.serve())

    def stop(self):
        if self.task and not self.task.done():
            self.task.cancel()
            print("API server stopped.")
        else:
            print("API server is not running or already stopped.")

# Create a stopper instance
stopper = ServerStopper(server)

# Run the server in a separate task using asyncio.create_task
# This allows the cell to continue executing after starting the server
# and prevents the RuntimeError previously encountered.

# To start the server, call:
# await stopper.start()
# To stop the server, call:
# stopper.stop()

# We will start it now for the user.
await stopper.start()
```

... INFO: Started server process [1995]
INFO: Waiting for application startup.
API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000

Task 2 – Library Book Management API

Task:

Develop a RESTful API to handle library books.

Instructions:

- Endpoints required:

- o GET /books → Retrieve all books

- o POST /books → Add a new book

- o GET /books/{id} → Get details of a specific book

- o PATCH /books/{id} → Update partial book details
(e.g., availability)

- o DELETE /books/{id} → Remove a book

- Implement error handling for invalid requests.

Expected Output:

- Functional API with CRUD + partial updates

```
Commands Code Text Run All
```

[1] ✓ 4s

```
pip install fastapi uvicorn
```

...

Requirement already satisfied: fastapi in /usr/local/lib/python3.12/dist-packages (0.135.1)
Requirement already satisfied: uvicorn in /usr/local/lib/python3.12/dist-packages (0.41.0)
Requirement already satisfied: starlette>=0.46.0 in /usr/local/lib/python3.12/dist-packages (from fastapi) (0.46.0)
Requirement already satisfied: pydantic>=2.7.0 in /usr/local/lib/python3.12/dist-packages (from fastapi) (2.10.6)
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from fastapi) (0.0.4)
Requirement already satisfied: click>=7.0 in /usr/local/lib/python3.12/dist-packages (from uvicorn) (8.1.8)
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.12/dist-packages (from uvicorn) (0.14.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic) (2.41.4)
Requirement already satisfied: anyio<5,>=3.6.2 in /usr/local/lib/python3.12/dist-packages (from anyio) (4.6.2)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio) (3.10)

I've installed `fastapi` and `uvicorn`. Now, let's define the API for managing library books. I'll use `pydantic` for data validation and `fastapi` for the API endpoints. The book records will be stored in a simple Python dictionary in memory.

```
[2] ✓ 0s
```

```
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
from typing import Dict, List, Optional

app = FastAPI(title="Library Book Management API")

# In-memory data store for books
books_db: Dict[int, BaseModel] = {}
next_book_id = 1

# Pydantic model for a book
class Book(BaseModel):
    title: str
    author: str
    publication_year: int
    is_available: bool = True
    id: Optional[int] = None # Optional ID for response, will be assigned by API

# Pydantic model for updating a book (all fields optional)
class BookUpdate(BaseModel):
    title: Optional[str] = None
    author: Optional[str] = None
    publication_year: Optional[int] = None
    is_available: Optional[bool] = None
```

[2]

✓ Os



```
@app.get("/books", response_model=List[Book], summary="Retrieve all books")
def get_all_books():
    """Retrieve a list of all books in the library."""
    return list(books_db.values())

@app.post("/books", response_model=Book, status_code=201, summary="Add a new book")
def add_book(book: Book):
    """Add a new book to the library.

    - **title**: The title of the book.
    - **author**: The author of the book.
    - **publication_year**: The year the book was published.
    - **is_available**: Whether the book is currently available (defaults to True).
    """
    global next_book_id
    book.id = next_book_id
    books_db[next_book_id] = book
    next_book_id += 1
    return book

@app.get("/books/{book_id}", response_model=Book, summary="Get details of a specific book")
def get_book_details(book_id: int):
    """Get details of a specific book by its ID."""
    if book_id not in books_db:
        raise HTTPException(status_code=404, detail="Book not found")
```

[2]

✓ Os



```
@app.patch("/books/{book_id}", response_model=Book, summary="Update partial book details")
def update_book_details(book_id: int, book_update: BookUpdate):
    """Update partial details of a book by its ID.

    - **title**: New title (optional).
    - **author**: New author (optional).
    - **publication_year**: New publication year (optional).
    - **is_available**: New availability status (optional).
    """
    if book_id not in books_db:
        raise HTTPException(status_code=404, detail="Book not found")

    existing_book = books_db[book_id]
    update_data = book_update.dict(exclude_unset=True) # Only update provided fields
    updated_book_data = existing_book.dict()
    updated_book_data.update(update_data)
    updated_book = Book(**updated_book_data)
    books_db[book_id] = updated_book
    return updated_book

@app.delete("/books/{book_id}", status_code=204, summary="Remove a book")
def delete_book(book_id: int):
    """Remove a book from the library by its ID."""
    if book_id not in books_db:
        raise HTTPException(status_code=404, detail="Book not found")
```

```
[2] ✓ Os del books_db[book_id]
return {"message": "Book deleted successfully"}
```

Now that the API is defined, you can run it using Uvicorn. Since you are in a Colab environment, we need to use `nest_asyncio` to allow Uvicorn to run within the existing event loop. Also, the API will be accessible via `localhost:8000`. You can test it by running the following cell.

To stop the API server later, you can run `stopper.stop()` in a new cell.

To interact with the API, you can use `curl` commands in a new cell or a separate terminal:

- **GET all books:**

```
!curl http://localhost:8000/books
```

- **POST a new book:**

```
!curl -X POST -H "Content-Type: application/json" -d '{"title": "The Hobbit", "author": "J.R.R.
```

- **GET a specific book (e.g., ID 1):**

```
!curl -X POST -H "Content-Type: application/json" -d '{"title": "The Hobbit", "author": "J.R.R.
```

- **GET a specific book (e.g., ID 1):**

```
!curl http://localhost:8000/books/1
```

- **PATCH (update) book details (e.g., change availability for ID 1):**

```
!curl -X PATCH -H "Content-Type: application/json" -d '{"is_available": false}' http://locall
```

- **DELETE a book (e.g., ID 1):**

```
[3] 0s
import uvicorn
import nest_asyncio
import asyncio

nest_asyncio.apply()

# Configuration for Uvicorn server
config = uvicorn.Config(app, host="0.0.0.0", port=8000, loop="asyncio")
server = uvicorn.Server(config)

# Define a stopper to gracefully stop the server
class ServerStopper:
    def __init__(self, server_instance):
        self.server = server_instance
        self.task = None

    async def start(self):
        print("Library Book Management API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000")
        self.task = asyncio.create_task(self.server.serve())

    def stop(self):
        if self.task and not self.task.done():
            self.task.cancel()
            print("API server stopped.")
        else:
            print("API server is not running or already stopped.")

def stop(self):
    if self.task and not self.task.done():
        self.task.cancel()
        print("API server stopped.")
    else:
        print("API server is not running or already stopped.")

# Create a stopper instance
stopper = ServerStopper(server)

# Start the server
await stopper.start()
```

... INFO: Started server process [3359]
INFO: Waiting for application startup.
Library Book Management API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000

+ Code + Text Add text cell

Task 3 – Employee Payroll API

Task:

Create an API for managing employees and their salaries.

Instructions:

- Endpoints required:
 - o GET /employees → List all employees
 - o POST /employees → Add a new employee with salary details
 - o PUT /employees/{id}/salary → Update salary of an

employee

o DELETE /employees/{id} → Remove employee

from system

- Use AI to:

- o Suggest data model structure.

- o Add comments/docstrings for all endpoints.

Expected Output:

- API supporting salary management with clear documentation.

The screenshot shows a code editor interface with a terminal window at the top and a sidebar on the left. The terminal window displays the command `pip install fastapi uvicorn` and its output, which lists various dependencies already installed in the environment. Below the terminal, a text box contains the message: "I've ensured `fastapi` and `uvicorn` are installed. Now, let's define the data models for the Employee Payroll API."

The sidebar on the left contains a search bar and a list of tabs: "Commands", "Code", "Text", and "Run all". The "Suggested Data Model Structure" tab is selected, showing a recommendation to use Pydantic models for data validation and serialization. It lists two models: `Employee` and `SalaryUpdate`. The `Employee` model is described as representing an employee with fields for name, position, and salary, where `id` is optional for input but assigned by the API. The `SalaryUpdate` model is described as handling updates to an employee's salary. It also states that employee records will be stored in an in-memory dictionary for simplicity.

Below the suggestions, a code editor shows the following Python code:

```
[2]
✓ 0s
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
from typing import Dict, List, Optional

app = FastAPI(title="Employee Payroll API")

# In-memory data store for employees
employees_db: Dict[int, BaseModel] = {}
next_employee_id = 1

# Pydantic model for an Employee
class Employee(BaseModel):
    name: str
    position: str
```

```

class Employee(BaseModel):
    name: str
    position: str
    salary: float
    id: Optional[int] = None # Optional ID for response, will be assigned by API

# Pydantic model for updating only the salary
class SalaryUpdate(BaseModel):
    salary: float

@app.get("/employees", response_model=List[Employee], summary="List all employees")
def list_employees():
    """Retrieve a list of all employees in the system, including their details."""
    return list(employees_db.values())

@app.post("/employees", response_model=Employee, status_code=201, summary="Add a new employee")
def add_employee(employee: Employee):
    """Add a new employee to the system with their name, position, and salary.

    - **name**: The full name of the employee.
    - **position**: The job title or position of the employee.
    - **salary**: The current salary of the employee.
    """

```

```

[2]
✓ 0s
global next_employee_id
employee.id = next_employee_id
employees_db[next_employee_id] = employee
next_employee_id += 1
return employee

@app.put("/employees/{employee_id}/salary", response_model=Employee, summary="Update salary of")
def update_employee_salary(employee_id: int, salary_update: SalaryUpdate):
    """Update the salary of a specific employee identified by their ID.

    - **employee_id**: The unique identifier of the employee.
    - **salary**: The new salary value for the employee.
    """
    if employee_id not in employees_db:
        raise HTTPException(status_code=404, detail="Employee not found")

    existing_employee = employees_db[employee_id]
    existing_employee.salary = salary_update.salary
    employees_db[employee_id] = existing_employee
    return existing_employee

@app.delete("/employees/{employee_id}", status_code=204, summary="Remove an employee")
def delete_employee(employee_id: int):
    """Remove an employee record from the system by their ID.

```

```
[2]
✓ 0s
"""
employee_id : the unique identifier of the employee to be removed.
"""
if employee_id not in employees_db:
    raise HTTPException(status_code=404, detail="Employee not found")
del employees_db[employee_id]
return {"message": "Employee deleted successfully"}
```

Now that the Employee Payroll API is defined, you can run it using Uvicorn. Since you are in a Colab environment, we need to use `nest_asyncio` to allow Uvicorn to run within the existing event loop. The API will be accessible via `localhost:8000`. You can test it by running the following cell.

To stop the API server later, you can run `stopper.stop()` in a new cell.

To interact with the API, you can use `curl` commands in a new cell or a separate terminal:

- GET all employees:

```
!curl http://localhost:8000/employees
```

- POST a new employee:

```
!curl -X POST -H "Content-Type: application/json" -d '{"name": "John Doe", "position": "Soft
```

- PUT (update) an employee's salary (e.g., for employee ID 1):

```
!curl -X PUT -H "Content-Type: application/json" -d '{"salary": 80000.0}' http://localhost:80
```

- DELETE an employee (e.g., employee ID 1):

```
!curl -X DELETE http://localhost:8000/employees/1
```

```
[3]
✓ 0s
import uvicorn
import nest_asyncio
import asyncio

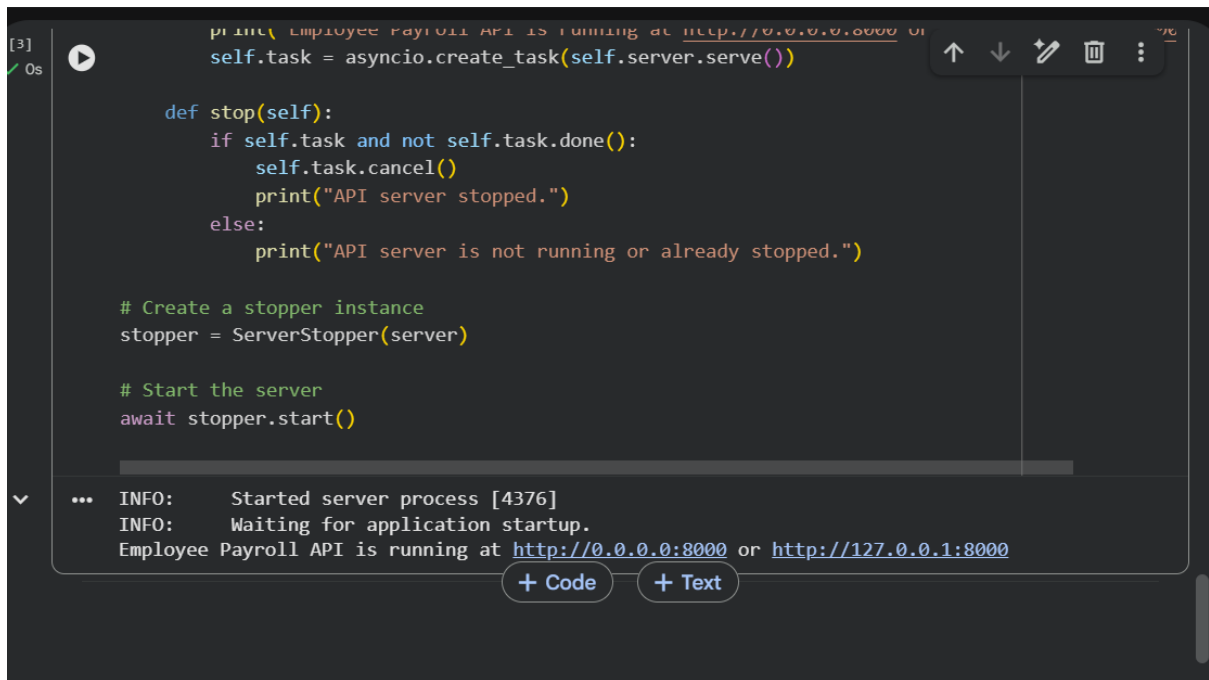
nest_asyncio.apply()

# Configuration for Uvicorn server
config = uvicorn.Config(app, host="0.0.0.0", port=8000, loop="asyncio")
server = uvicorn.Server(config)

# Define a stopper to gracefully stop the server
class ServerStopper:
    def __init__(self, server_instance):
        self.server = server_instance
        self.task = None

    async def start(self):
        print("Employee Payroll API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000")
        self.task = asyncio.create_task(self.server.serve())

    def stop(self):
        if self.task and not self.task.done():
            self.task.cancel()
            print("API server stopped.")
        else:
```



```
[3] 0s
print("Employee Payroll API is running at http://0.0.0.0:8000 or")
self.task = asyncio.create_task(self.server.serve())

def stop(self):
    if self.task and not self.task.done():
        self.task.cancel()
        print("API server stopped.")
    else:
        print("API server is not running or already stopped.")

# Create a stopper instance
stopper = ServerStopper(server)

# Start the server
await stopper.start()
```

... INFO: Started server process [4376]
INFO: Waiting for application startup.
Employee Payroll API is running at <http://0.0.0.0:8000> or <http://127.0.0.1:8000>

+ Code + Text

Task 4 – Real-Time Application: Online Food Ordering API

Scenario:

Design a simple API for an online food ordering system.

Requirements:

- Endpoints required:
 - o GET /menu → List available dishes
 - o POST /order → Place a new order
 - o GET /order/{id} → Track order status
 - o PUT /order/{id} → Update an existing order (e.g., change items)
 - o DELETE /order/{id} → Cancel an order
- AI should generate:
 - o REST API code
 - o Suggested improvements (like authentication, pagination)

Expected Output:

- Fully working API simulating a food ordering backend.

Deliverables (For All Tasks)

1. AI-generated prompts for code and test case generation.
2. At least 3 assert test cases for each task.

3. AI-generated initial code and execution screenshots.
4. Analysis of whether code passes all tests.
5. Improved final version with inline comments and explanation.
6. Compiled report (Word/PDF) with prompts, test cases, assertions, code, and output.

```
[1] 6s pip install fastapi uvicorn

... Requirement already satisfied: fastapi in /usr/local/lib/python3.12/dist-packages (0.135.1)
Requirement already satisfied: uvicorn in /usr/local/lib/python3.12/dist-packages (0.41.0)
Requirement already satisfied: starlette>=0.46.0 in /usr/local/lib/python3.12/dist-packages (from fastapi) (0.46.0)
Requirement already satisfied: pydantic>=2.7.0 in /usr/local/lib/python3.12/dist-packages (from fastapi) (2.10.6)
Requirement already satisfied: typing-extensions>=4.8.0 in /usr/local/lib/python3.12/dist-packages (from pydantic) (4.12.2)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic) (0.4.2)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from fastapi) (0.0.2)
Requirement already satisfied: click>=7.0 in /usr/local/lib/python3.12/dist-packages (from uvicorn) (8.1.8)
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.12/dist-packages (from uvicorn) (0.14.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic) (0.6.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic) (2.41.4)
Requirement already satisfied: anyio<5,>=3.6.2 in /usr/local/lib/python3.12/dist-packages (from uvicorn) (4.6.2)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio) (3.10)

I've ensured fastapi and uvicorn are installed. Now, let's define the API for the online food ordering system. I'll use pydantic for data validation and fastapi for the API endpoints. Menu items and orders will be stored in in-memory dictionaries.
```

```
[2] 0s from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
from typing import Dict, List, Optional
import uuid

app = FastAPI(title="Online Food Ordering API")

# In-memory data store for menu items and orders
menu_db: Dict[str, BaseModel] = {}
orders_db: Dict[str, BaseModel] = {}

# Initialize some sample menu items
menu_db["dish1"] = {"name": "Pizza Margherita", "description": "Classic pizza with tomato and mozzarella"}
menu_db["dish2"] = {"name": "Spaghetti Carbonara", "description": "Pasta with eggs, hard cheese, and pancetta"}
menu_db["dish3"] = {"name": "Caesar Salad", "description": "Fresh romaine lettuce with Caesar dressing"}

# Pydantic model for a Dish
class Dish(BaseModel):
    name: str
    description: str
    price: float
    dish_id: Optional[str] = None # Added for clarity, will be key in menu_db

# Pydantic model for an item within an Order
class OrderItem(BaseModel):
```

```

# Pydantic model for an item within an Order
class OrderItem(BaseModel):
    dish_id: str
    quantity: int

# Pydantic model for an Order
class Order(BaseModel):
    items: List[OrderItem]
    status: str = "pending"
    total_price: float = 0.0
    order_id: Optional[str] = None # Will be generated by the API

# Pydantic model for updating an existing order
class OrderUpdate(BaseModel):
    items: Optional[List[OrderItem]] = None
    status: Optional[str] = None

@app.get("/menu", response_model=List[Dish], summary="List available dishes")
def get_menu():
    """Retrieve a list of all available dishes on the menu."""
    # Convert menu_db values to Dish models and add dish_id
    dishes = []
    for dish_id, dish_data in menu_db.items():
        dish_obj = Dish(**dish_data)

```

```

[2]
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        dishes.append(dish_obj)
    return dishes

@app.post("/order", response_model=Order, status_code=201, summary="Place a new order")
def place_order(order: Order):
    """Place a new order with a list of dishes and quantities.

    - **items**: A list of `OrderItem` objects, each containing `dish_id` and `quantity`.
    """
    order_id = str(uuid.uuid4()) # Generate a unique order ID
    total_price = 0.0

    # Calculate total price and validate dishes
    for item in order.items:
        if item.dish_id not in menu_db:
            raise HTTPException(status_code=404, detail=f"Dish '{item.dish_id}' not found in menu")
        dish_price = menu_db[item.dish_id]["price"]
        total_price += dish_price * item.quantity

    order.order_id = order_id
    order.total_price = total_price
    orders_db[order_id] = order
    return order

@app.get("/order/{order_id}", response_model=Order, summary="Track order status")

```

```

2] def get_order_status(order_id: str):
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    """Track the status and details of a specific order by its ID."""
    if order_id not in orders_db:
        raise HTTPException(status_code=404, detail="Order not found")
    return orders_db[order_id]

@app.put("/order/{order_id}", response_model=Order, summary="Update an existing order")
def update_order(order_id: str, order_update: OrderUpdate):
    """Update an existing order's items or status.
    Note: This allows for significant changes, in a real app this might be more restricted.

    - **items**: An optional list of `OrderItem` objects to replace current items.
    - **status**: An optional string to update the order status (e.g., 'preparing', 'delivered')
    """
    if order_id not in orders_db:
        raise HTTPException(status_code=404, detail="Order not found")

    existing_order = orders_db[order_id]
    update_data = order_update.dict(exclude_unset=True)

    # Update items if provided
    if "items" in update_data:
        new_total_price = 0.0
        for item in update_data["items"]:
            if item["dish_id"] not in menu_db:
                raise HTTPException(status_code=404, detail=f"Dish '{item['dish_id']}' not found")

```

```

        if item["dish_id"] not in menu_db:
            raise HTTPException(status_code=404, detail=f"Dish '{item['dish_id']}' not found")
        dish_price = menu_db[item["dish_id"]]["price"]
        new_total_price += dish_price * item["quantity"]
        existing_order.items = [OrderItem(**item) for item in update_data["items"]]
        existing_order.total_price = new_total_price

    # Update status if provided
    if "status" in update_data:
        existing_order.status = update_data["status"]

    orders_db[order_id] = existing_order
    return existing_order

@app.delete("/order/{order_id}", status_code=204, summary="Cancel an order")
def cancel_order(order_id: str):
    """Cancel an existing order by its ID."""
    if order_id not in orders_db:
        raise HTTPException(status_code=404, detail="Order not found")

    # In a real system, you might change status to 'cancelled' instead of deleting
    del orders_db[order_id]
    return {"message": "Order cancelled successfully"}

```

Now that the Online Food Ordering API is defined, you can run it using Uvicorn. The API will be accessible via `localhost:8000`. You can test it by running the following cell.

To stop the API server later, you can run `stopper.stop()` in a new cell.

To interact with the API, you can use `curl` commands in a new cell or a separate terminal. Note that `order_id`s are UUIDs and will be generated by the `POST /order` endpoint.

- **GET menu:**

```
!curl http://localhost:8000/menu
```

- **POST a new order:**

```
!curl -X POST -H "Content-Type: application/json" -d '{"dish_id": "dish1", "quantity": 2},
```

(Replace `dish_id` with actual IDs from `/menu`)

- **GET order status (replace `YOUR_ORDER_ID`):**

```
!curl http://localhost:8000/order/YOUR_ORDER_ID
```

[3]
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```
import uvicorn
import nest_asyncio
import asyncio

nest_asyncio.apply()

# Configuration for Uvicorn server
config = uvicorn.Config(app, host="0.0.0.0", port=8000, loop="asyncio")
server = uvicorn.Server(config)

# Define a stopper to gracefully stop the server
class ServerStopper:
    def __init__(self, server_instance):
        self.server = server_instance
        self.task = None

    async def start(self):
        print("Online Food Ordering API is running at http://0.0.0.0:8000 or http://127.0.0.1:")
        self.task = asyncio.create_task(self.server.serve())

    def stop(self):
        if self.task and not self.task.done():
```

```
3] 0s ▶ self.server = server_instance
    self.task = None

    async def start(self):
        print("Online Food Ordering API is running at http://0.0.0.0:8000 or http://127.0.0.1:8000")
        self.task = asyncio.create_task(self.server.serve())

    def stop(self):
        if self.task and not self.task.done():
            self.task.cancel()
            print("API server stopped.")
        else:
            print("API server is not running or already stopped.")

    # Create a stopper instance
    stopper = ServerStopper(server)

    # Start the server
    await stopper.start()
```

... INFO: Started server process [5333]
INFO: Waiting for application startup.
Online Food Ordering API is running at <http://0.0.0.0:8000> or <http://127.0.0.1:8000>